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Started	Tuesday, 20 October 2026, 4:21 PM
Completed	Tuesday, 20 October 2026, 4:32 PM
Duration	10 mins 34 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By dividing the list into halves
c. By comparing each element in the list sequentially ✓
d. By sorting the list first

Question **2**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the search algorithm finishes
b. When the list contains duplicate elements
c. When the element is found in the list ✓
d. When the list is sorted

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By dividing the list length by two ✓
c. By using a hash function
d. By comparing each element sequentially

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. There must be a mechanism to delete and/or insert elements in the list ✓
- b. The list must be sorted
- c. There should be direct access to the middle element in any sublist
- d. Number values should only be present

Number values should only be present

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It does not work with negative numbers
- b. It is slower than linear search for small lists
- c. It requires the list to be sorted ✓
- d. It can only be applied to large lists

Question **6**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Hash
- b. Binary ✓
- c. Linear
- d. Both (1) & (3)

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Dynamic search
- b. Hashing
- c. Linear search ✓
- d. Binary search

Question **8**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Hash search
- b. Binary search ✓
- c. Linear search
- d. Sequential search

Question **9**

Correct

Mark 1.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The middle element is chosen randomly
- b. The search stops
- c. The higher middle element is chosen as the middle element ✓
- d. The lower middle element is chosen as the middle element

Question **10**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O👎
- b. O(log n) ✓
- c. O(1)
- d. O(n log n)

Question **11**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

- a. Divide the list into two equal parts
- b. Compare the target element with the middle element in the list
- c. Compare the target element with the first element in the list
- d. Sort the list ✗

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. There must be a mechanism to access middle element directly
- b. Must use a sorted array
- c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is unsorted
- b. When the list is sorted ✓
- c. When the list is very small
- d. When the list contains duplicate elements

Question **14**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 500
- b. 100
- c. 10 ✓
- d. 1000

Question **15**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Jump search
c. Linear search ✓
d. Binary search